Tectonophysics, 64 (1980) 163—172

© Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

PALEOMAGNETIC STUDIES COMBINED WITH FISSION-TRACK DATINGS ON THE WESTERN ARC OF SULAWESI, EAST INDONESIA

SADAO SASAJIMA [1], SUSUMU NISHIMURA [2], KIMIO HIROOKA [3], YOICHIRO OTOFUJI [1], THEO VAN LEEUWEN [4] and FRED HEHUWAT [5]

[1] *Department of Geology and Mineralogy, Kyoto University, 606 Sakyo Kyoto (Japan)*
[2] *Institute of Earth Science, Kyoto University, 606 Sakyo Kyoto (Japan)*
[3] *Department of Earth Sciences, Toyama University, 930 Gohuku, Toyama (Japan)*
[4] *P.T. Rio Tinto Indonesia, Cilandak Commercial Estate Bld. 411, Cilandak, Kebayoran Timur, Jakarta (Indonesia)*
[5] *National Institute of Geology and Mining-LIPI, Jl. Cisitu No. 21, Bandung (Indonesia)*

(Received November 24, 1978; accepted for publication January 18, 1979)

ABSTRACT

Sasajima, S., Nishimura, S., Hirooka, K., Otofuji, Y., Van Leeuwen, Th. and Hehuwat, F., 1979. Paleomagnetic studies combined with fission-track datings on the western arc of Sulawesi, East Indonesia. Tectonophysics, 64: 163—172.

Paleomagnetic studies in conjunction with fission-track dating on the western arc of Sulawesi yield important evidence bearing on the tectonic history of the area. During the Paleogene to Early Miocene time interval the paleomagnetic pole for southwestern Sulawesi was situated at 36.5°E 44.8°N. This pole position is significantly different from that in the time interval Middle Miocene to Recent, which is consistent with the north pole of the axial geocentric dipole. This fact suggests that subsequent to the Paleogene to Early Miocene period, possibly 19—13 m.y. B.P., a major tectonic event occurred which caused about 40 degrees of anticlockwise rotation of the area. It is suggested by the present work that the postulated collision followed by welding of eastern Sulawesi with western Sulawesi during the Pliocene (Katili, 1978) may be the tectonic event mentioned above. In addition, our data does not support the hypothesis that western Sulawesi has been derived from the dispersal of Gondwanaland.

INTRODUCTION

Sulawesi consists of two major tectonic units, namely an eastern arc, dominated by thrust tectonics in connection with emplacement of an extensive ophiolite sequence, and a western arc, characterized by normal folding, widespread Cenozoic volcanism and granite intrusions.

Many hypotheses on the tectonic history of Sulawesi have so far been put forward to account for the complex geological and geophysical features of the island. The most noteworthy are those by Katili (1975, 1978), Audley-Charles et al. (1972) and Audley-Charles (1974). The authors have suggested in these publications that their ideas should be confirmed by paleomagnetic

methods before being considered established. Based on a reconnaissance paleomagnetic survey in southwest and southeast Sulawesi, Haile (1978) found evidence to support quite separate paleogeographic positions of these two regions at the Early Cretaceous period, as postulated originally by Audley-Charles et al. (1972).

One of the main purposes of our studies is to offer a tectonic framework for the western arc of Sulawesi from the paleomagnetic view point. Reconnaissance paleomagnetic investigations combined with fission-track age dating were carried out by us in southwest Sulawesi (Biru area), central Sulawesi (Gimpu and Megapa) and north Sulawesi (Kampong Kombi and Tansealama). Sample localities are shown in Fig. 1. The geologic setting of these three regions will be described in the following section. Most rock samples (eighteen sites) were collected from the Biru area, as for this area a detailed geological map was available. As can be seen from Fig. 1, three of the sample localities occur close two prominent fault zones, the Watampone fault and Palu—Koro strike-slip fault (Tjia and Zakaria, 1974), and therefore special care was taken to collect undisturbed rock samples.

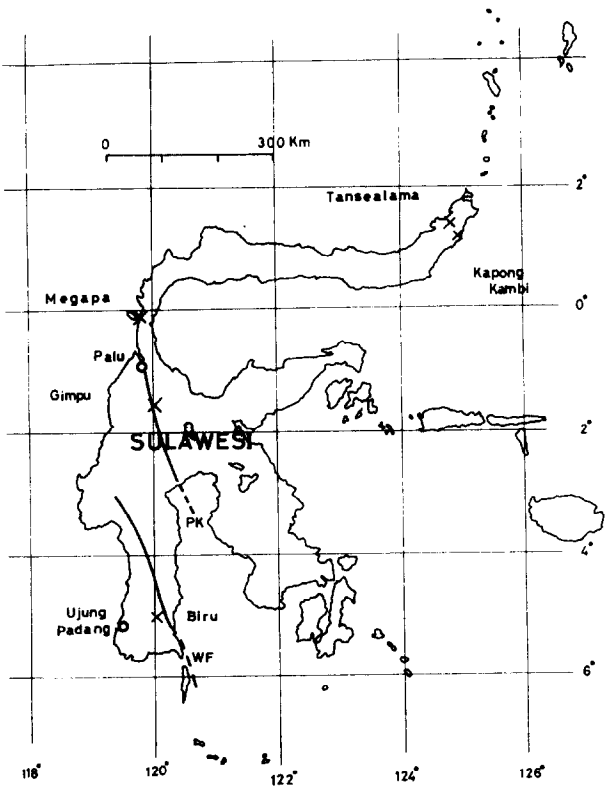


Fig. 1. Map of Sulawesi showing the sample localities. x = sampling sites; *PK* = Palu—Koro fault; *WF* = Watampone fault.

GEOLOGIC SETTING

Southwest Sulawesi

Regional mapping of south Sulawesi has been carried out by the Geological Survey of Indonesia (Sukamto, 1975). The Biru area was investigated in detail during a mineral survey by Rio Tinto Indonesia (Van Leeuwen, 1975). Its generalized stratigraphy and the sampling horizons are shown in Table I. The oldest rocks of South Sulawesi are basement complexes of polymetamorphic rocks, which are unconformably overlain by Lower Cretaceous deep-sea sediments. These in turn are unconformably overlain by a series of Upper Cretaceous shale, greywacke and arkose with interbedded volcanics (Marada Sandstone).

After a period of folding and erosion, a NNW trending calc-alkaline volcanic chain (Langi volcanics) developed during the Paleocene. Towards the

TABLE I

Stratigraphic sequence of sampling sites in the Biru area

Geological age	Thickness (m)	Lithological unit	Fission track age	Paleomagnetic sampling site *
Pliocene	350+	Lemo Volcanics		ID 114, ID 115
		— — — — — — — — — — —		
Upper to Middle Miocene	250	Pamessurang Volcanic Complex	13 m.y. (ID 116)	ID 116, ID 117 ID 118,
		— — — — — — — — — — —		
		Intrusion of Granodiorite and Quartzdiorite	19 m.y. (ID 108)	ID 101, ID 108 ID 109, ID 110
		— — — — — — — — — — —		
Middle Miocene	200+	Upper Tonassa Limestone		
		— — — — — — — — — — —		
Upper Eocene	100—250	Lower Tonassa Limestone		ID 103
		———————————		
Paleocene	400+	Langi Volcanics	63 m.y. (≃ID 105)	ID 102, ID 104 ID 105, ID 106 ID 107
		relation uncertain (not exposed)		
Upper Cretaceous	1000	Marada Sandstone		ID 111, ID 112 ID 113

* : Rock kind of the sample can be referred to Table III.
≃: Sample for fission track dating was taken from the intercalated tuff layer within andesite lavas; its sampling site was near the paleomagnetic sampling site ID 105.
— — — : Unconformity.
——— : Conformity.

end of the Eocene volcanic activity ceased and a long period of carbonate deposition began (Tonassa Limestone). At the beginning of the Middle Miocene a major orogenic event took place, which was accompanied by andesitic volcanism and granite intrusions. As a result, the Lower Tertiary sequence was folded, faulted and strongly uplifted. This orogenic event was followed by a long period of volcanic activity during which huge amounts of alkaline and calc-alkaline volcanic rocks were deposited on land and in shallow marine basins (Pamessurang Volcanics, Lemo Volcanics).

A second major orogenic event took place during Plio-Pleistocene times, lifting practically the whole of southwest Sulawesi above sea level. Volcanic activity continued until the young Quaternary during which time it almost completely ceased.

Central Sulawesi

Central Sulawesi is largely occupied by a thick sequence of flysh-like and other marine sediments with interbedded calc-alkaline volcanics (Tinombo Formation), which rests unconformably on a metamorphic basement. The rocks were deposited in basins flanking the Langi Volcanics to the northeast. To date only Eocene guide fossils have been found in this formation, but its lower part is possibly Cretaceous in age. The rocks of the Tinombo Formation were strongly folded during the Oligocene or Middle Miocene. They are unconformably overlain by Miocene shallow marine sediments (Celebes Molasse). Plio-Pleistocene tuffs are locally present along the Palu—Koro fault zone, which is a major sinistral strike-slip fault. Limited radiometric data suggest that intrusive rocks were emplaced during the Oligocene and Pliocene. As in Southwest Sulawesi the region was strongly uplifted during the Plio-Pleistocene orogenic event.

North Sulawesi

According to recent investigations by P.T. Tropic Endeavour (Trial et al., 1974), the oldest formation (Cretaceous—Paleogene) comprises a thick sequence of basaltic rocks (pillow lava, spillite, etc.) with interlayered deep-sea sediments, which locally are intruded by swarms of basic dykes. On this oceanic basement a chain of intermediate and acid calc-alkaline volcanoes (Bilunga Volcanics) was built during the Miocene. Contemporaneous volcano-clastic and normal sediments, pyroclastics and lava flows were deposited in shallow basins flanking the volcanic chain. Renewed volcanic activity in the Plio-Pleistocene produced acid and intermediate volcanics (predominantly pyroclastics) along the entire length of north Sulawesi. Young strato-volcanoes, some still active, are present in the northeastern part. Intrusive rocks are widespread throughout the western and central parts of north Sulawesi. Large masses of predominantly dioritic intrusive rock (Middle to Upper Miocene) are intruded into the Bilungan Volcanics and associated rocks. Younger, more acid intrusives are also present.

RESULTS

Several samples from acid intrusive rocks and tuffs from different stratigraphic horizons were subjected to age determination by the fission-track dating technique. Ages for a few samples which contained considerable zircon crystals with rich uranium concentration have been determined. For more detailed description of the dating method the reader is referred to the previous paper by Nishimura and Yokoyama (1973). The results so far obtained are listed in Table II and also in Table I.

The remanent magnetizations of the specimen before and after stepwise a.f. demagnetization treatment were measured by means of a spinner

TABLE II

Fission-track age of some rocks; mean of several numbers of zircon crystals

Sample	No. of zircons	Spontaneous fission-track density ρ_s (cm^{-2})	Induced fission-track density ρ_i (cm^{-2})	Fission-track age T (m.y.)
ID 105	1	$7.47 \cdot 10^6$	$5.15 \cdot 10^6$	62
	2	6.61	4.35	65
	3	5.11	3.57	61
	4	6.12	4.08	64
	5	7.25	4.71	66
	6	5.67	4.05	60
				63 ± 2.2
Neutron dose $\Phi = 0.70 \cdot 10^{15}$ (cm^{-2})				
ID 108	1	$1.65 \cdot 10^6$	$4.25 \cdot 10^6$	25
	2	1.08	3.86	18
	3	1.21	5.21	15
	4	1.11	3.56	20
	5	1.20	5.15	15
	6	1.50	5.07	19
				19 ± 3.4
Neutron dose $\Phi = 1.05 \cdot 10^{15}$ (cm^{-2})				
ID 116	1	$1.63 \cdot 10^6$	$3.88 \cdot 10^6$	18
	2	1.43	5.10	12
	3	1.07	4.57	10
	4	1.10	3.92	12
	5	1.60	5.25	13
	6	2.20	5.89	16
	7	0.89	3.78	10
				13 ± 2.8

Neutron dose $\Phi = 0.70 \cdot 10^{15}$ (cm^{-2})

Age (m.y.) $T = 6.12 \cdot 10^{-14}\ \Phi \cdot \rho_s/\rho_i$

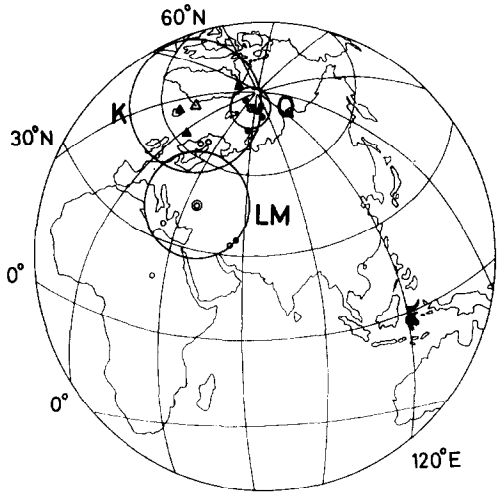


Fig. 2. Plaeomagnetic poles derived from the western arc of Sulawesi since Late Cretaceous. *K* = Late Cretaceous; *LM* = Paleocene to Early Miocene; *Q* = Middle Miocene to Recent.

magnetometer (Schonstedt SSM-1A). After remanent stability tests on the pilot specimens about half of the samples were discarded from further paleomagnetic research owing to their unreliable nature. The paleomagnetic properties and the related data are summarized in Table III. The paleomagnetic poles with their circles of 95% confidence level of the Upper Cretaceous, Paleocene to Lower Miocene and Middle Miocene to Recent are respectively shown in Fig. 2.

DISCUSSION

From Table III and Fig. 2 it appears that since the Middle Miocene (13 m.y. B.P.) to Recent all the VGP's significantly group about the present north pole, suggesting that the paleomagnetic pole position is consistent with the present north pole within the range of errors. Contrary to this, the VGP's for pre-Middle Miocene rocks show a fairly scattered distribution. It may be very difficult to derive a reliable unique explanation of the paleomagnetic features based on these scattered groupings of VGP's. Needless to say, such a divergence of the VGP's cannot be reconciled merely with polar wandering and paleo-secular variation for the period concerned. However, the data might be explained by the occurrence of a major tectonic event resulting in a fairly large anticlockwise rotation (—45°) of southwest Sulawesi during the interval ranging from 63 to 13 m.y., but most probably 19—13 m.y. B.P. In the Biru area this event is represented by a pronounced angular unconformity between folded rocks of Paleocene to basal Middle Miocene age and flat-lying Neogene volcanics.

TABLE III

Summary of paleomagnetic results of the western arc of Sulawesi

Geological age	Site	Rock type	N	a.f. peak field	D	I	α_{95}	Pole position	
								Long.	Lat. (°N)
Recent	Tansealama	Tuff	7	100	−13.2	10.0	4.5	50.9°E	76.3
Pliocene	Kampong Kombi	Ignimbrite	9	100	−5.1	15.1	2.9	86.8°E	81.8
	ID 114	Tuff	11	100	0.6	−24.1	6.1	64.4°E	82.4
Miocene	Megapa	Granite	9	200	−6.4	11.1	17.2	72.1°E	81.4
Upper to Middle Miocene	ID 118 *	Basalt	13	200	−43.6	−26.4	7.6	15.0°E	46.2
	ID 116	Ignimbrite	11	100	−2.1	−10.9	3.3	16.5°E	87.8
	mean				−5.1	−5.2	6.8	54.9°E	84.4
Lower Miocene	ID 101	Quartzdiorite	9	200	127.3	−32.8	3.4	55.2°E	33.2
	ID 108	Granodiorite	9	200	−21.9	−13.3	6.4	24.4°E	68.1
	ID 109	Granodiorite	9	300	−67.3	−6.7	12.2	28.5°E	22.9
	ID 110 *	Granodiorite	8	400	−14.3	58.3	19.9	104.5°E	44.1
Paleocene	ID 102	Andesite	11	200	129.6	−35.4	2.7	58.0°E	34.7
	ID 104 *	Andesite	6	200	116.7	67.1	7.7	21.9°W	20.8
	ID 105	Andesite	11	200	−53.9	−20.7	3.8	20.4°E	36.4
	ID 106 *	Andesite	8	200	−13.0	63.0	6.4	108.0°E	39.2
	ID 107	Andesite	9	200	−23.0	−19.5	10.6	16.2°E	66.7
	mean				−44.8	1.3	19.6	36.5°E	44.8
Upper Cretaceous	Gimpu	Sandstone	8	200	164.1	36.2	6.5	20.8°W	65.6
	ID 112	Shale	10	300	−26.1	−36.2	7.8	1.1°E	61.5
	ID 113	Sandstone	11	200	−0.5	−20.2	8.1	54.8°E	84.6
	mean				−14.2	−32.3	21.7	13.2°W	71.3

Location: Transealama: 124°50′E 1°21′N; Kampong Kombi: 124°57′E 1°15′N; Megapa: 119°52′E 0°10′S; Gimpu: 120°04′E 1°42′S; Biru (ID): 120°10′E 5°02′S.

N = no. of cleaned specimens; D = declination east of geomagnetic north; I = inclination positive downwards; α_{95} = radius of circle of 95% confidence.

* Refer to the text.

Another reason for the poorer grouping of the VGP's can be explained by extensive block movements for which tilting corrections were not possible because of the igneous nature of the samples. A large number of NW-trending faults and their conjugate minor faults, which exist in the Biru area have caused extensive differential block-faulting. If we assume that the scatter of VGP's are mostly resulting from the block movements and the paleosecular variation, we can regard the average VGP's as representative of the paleomagnetic pole position during the period.

As shown in Fig. 2 the paleomagnetic pole for the Paleocene to Early Miocene period is located adjacent to the Cretaceous paleomagnetic pole position for the Malaya Peninsula (35°E, 44°N: McElhinny et al., 1974). The VGP's of ID 104, 106 and 110 have been disregarded in calculating the paleomagnetic pole as they are assumed to indicate the transitional poles during polarity reversal.

It is interesting to note Haile's (1978) recent report that the Cretaceous paleomagnetic poles for the Malaya Peninsula and West Kalimantan (21°E 41°N: Haile et al., 1977), and the Jurassic—Early Cretaceous paleomagnetic pole for southwest Sulawesi (32°E 55°N) show a good agreement with each other. One of his important conclusions is that the three regions have formed part of the same plate which has rotated anticlockwise 35°—50° since the Cretaceous. However, he has not shown paleomagnetically, the time of occurrence of the anticlockwise rotations. The present study shows that the southwest-arm of Sulawesi could have been rotated during the time ranging from 63 to 13 m.y., most probably in the period, 19–13 m.y. B.P.

Very recently Katili (1978) has proposed a sophisticated model of the tectonic evolution of Sulawesi since the Miocene, with emphasis on the characteristics of mobile zone tectonics. He has discussed the past and present geotectonic position of Sulawesi by elaborating on his previous idea of the westward thrust of Sulawesi in the direction of the Asian continent (Katili, 1975).

We suggest that the Pliocene welding proposed by Katili (1978) of the western and eastern arcs of Sulawesi took place between 19—13 m.y. B.P., that is Early—Middle Miocene, and not during the Pliocene. In addition, our limited data does not show any evidence of the bending of the north arm of Sulawesi during the welding (Katili, 1978), it might be tentatively proposed that the anticlockwise rotation (about 45°) of the southwest-arm of Sulawesi during the period 19—13 m.y. B.P. was caused by the anticlockwise rotation of New Guinea (Green and Pitt, 1967) and/or the northwestward drift of New Guinea that was proposed by Andley-Charles et al. (1972).

Our Cretaceous VGP's are too few to average out the paleo-secular variation of the period, but when we assume the mean VGP's to represent the paleomagnetic pole, it is significantly distant from the three Cretaceous paleomagnetic poles mentioned above. We have no convincing explanation of the discrepancy. Discussion on the discrepancy of two Cretaceous paleomagnetic poles obtained by both Haile (1978) and the present authors from the

southwest-arm of Sulawesi should await further paleomagnetic investigations.

CONCLUSIONS

If we accept the tectonic evolution proposed by Katili, it may be better to regard his proposed welding time of Pliocene as Early to Middle Miocene (19—13 m.y.). By this change the intense tectonic movement is more readily accounted for by the proposed anticlockwise rotation of New Guinea during the Miocene (Green and Pitt, 1967) and/or the rapid northwestward movement of New Guinea (Audley-Charles et al., 1972).

Furthermore, our paleomagnetic results from Middle Miocene to Recent do not support the hypothesis of the bending of the northern part of the western arc of Sulawesi during and subsequent to the welding of the eastern arc of Sulawesi in the Pliocene (Katili, 1978).

The Cretaceous paleomagnetic poles from southwest Sulawesi obtained by Haile (1978) and the present authors are not consistent: however, the hypothesis of the Gondwana origin of the western arc of Sulawesi (Ridd, 1971; Audley-Charles et al., 1972) cannot be supported by either of them.

ACKNOWLEDGEMENTS

We wish to acknowledge the other team members of the survey, Dr. K. Ishizaka, Messrs. Y. Masuda and A. Dahrma for their assistance in the rock sampling.

REFERENCES

Audley-Charles, M.G., 1974. Sulawesi, Mesozoic—Cenozoic orogenic belt, data for orogenic studies. Geol. Soc. London, Spec. Publ., 4: 365—378.

Audley-Charles, M.G., Carter, D.J. and Milsom, J.S., 1972. Tectonic development of Eastern Indonesia in relation to Gondwana dispersal. Nature (London), Phys. Sci., 239: 35—39.

Green, R. and Pitt, R.P.B., 1967. Suggested rotation of New Guinea. J. Geomagn. Geoelectr., 19: 317—321.

Haile, N.S., 1978. Reconnaissance paleomagnetic results from Sulawesi, Indonesia, and their bearing on paleogeographic reconstructions. Tectonophysics, 46: 77—85.

Haile, N.S., McElhinny, M.W. and McDougall, I., 1977. Paleomagnetic data and radiometric ages from the Cretaceous of West Kalimantan, and their significance in interpreting regional structure. J. Geol. Soc., 133: 133—144.

Katili, J.A., 1975. Volcanism and plate tectonics in the Indonesian island. Tectonophysics, 26: 165—188.

Katili, J.A., 1978. Past and present geotectonic position of Sulawesi, Indonesia. Tectonophysics, 45: 289—322.

McElhinny, M.W., Haile, N.S. and Crawford, A.S., 1974. Paleomagnetic evidence shows Malay Peninsula was not a part of Gondwana. Nature, 252: 641—645.

Nishimura, S. and Yokoyama, T., 1973. Fission-track ages of volcanic ashes in 200 m core sample of Lake Biwa, Japan. Proc. Jpn. Acad., 49: 615—618.

Ridd, M.F., 1971. South-East Asia as a part of Gondwanaland. Nature, 234: 531—533.
Sukamto, R., 1975. Geologic map of Indonesia. Sheet VIII: Ujung Pandang, Scale 1 : 1,000,000. Geol. Surv. Indonesia.
Tjia, H.D. and Zakaria, T., 1974. Palu—Koro strike-slip fault zone, Central Sulawesi, Indonesia. Sains Malaysiana, 3 (1): 65—86.
Trail, D.S., John, T.V., Bird, M.C., Obial, R.C., Pertzel, B.A., Abiog, D.B. and Parwoto, S., 1974. The general geological survey of block 2, Sulawesi Utara, Indonesia. P.T. Tropic Endeavour Indonesia (unpublished report).
Van Leeuwen, Th. M., 1975. Geological investigation of KP 190 (Biru), South Sulawesi. P.T. Rio Tinto Indonesia (unpublished report).